

Difference Counter for In and Out Gates

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Sometimes, we need to know whether someone is still inside a building before it is locked. And if so, how many persons are still inside? There can be many other similar situations for which the circuit presented here can come handy. But it needs separate in and out gates to work and can count up to nine persons (inside) only. Though, it can be extended to count more.

Circuit and working

Fig. 1 shows circuit of the difference counter, built around four ICs, for use between entrance and exit doors. Switch S1, which is an SPDT switch, together with gates N1 and N2 of CD4011 (IC4) generates a pulse for incrementing counter CD40192 (IC1). CD40192 is a BCD pre-settable up/down (bidirectional) counter. The counter moves up or down with the positive clock edges applied to its corresponding pin 5 (clock up) or pin 4 (clock down).

When you press S1, its pole comes in contact with its terminal 3, which is connected to pin 5 of IC4. When you



release S1, its pole returns to its original position at terminal 1. Thus, for every action of pressing and releasing switch S1, there is an increment in display DIS1 till 9. Pressing switch S3 momentarily resets the counter to 0.

Switch S2 is also an SPDT switch that works exactly like switch S1, but it is used for decrementing the count in DIS1. Switch S2, together with gates N3 and N4 of IC4 generates a pulse for decrementing the counter in IC1.

The counter can be expanded to display more digits through the output pins 12 and 13 of IC1, if needed. High-level signal on pin 14 (reset) clears all internal triggers. That is, when switch S3 is pressed momentarily, IC1 is reset.

Test Points

Test point	Details
TP0	0V
TP1	5V
TP2	Low to high signal when S2 is pressed and released
TP3	Low to high signal when S1 is pressed and released
TP4-TP7	Binary values (0 or 1) equivalent to the respective displays shown on DIS1

Also, resetting occurs through capacitor C7 whenever power is switched on.

The outputs of IC1 are decoded by CD4511 (IC2), which has internal 4-bit latch, decoder and output drivers. The internal latch is always enabled because pin 5 (LE) is connected to ground permanently. The outputs of IC2 drive common-cathode 7-segment display LTS543 (DIS1). Since pin configurations may be different for the other displays, always check their datasheets

PARTS LIST

Semiconductors:	
IC1	- CD40192 up/down decade counter
IC2	- CD4511 BCD to 7-segment driver
IC3	- 7805, 5V voltage regulator
IC4	- CD4011 quad NAND gate
D1	- 1N4001 rectifier diode
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1-R4	- 10-kilo-ohm
R5-R11	- 330-ohm
R12	- 22-kilo-ohm
R13	- 1-kilo-ohm
Capacitors:	
C1-C3, C5, C7	- 0.33µF ceramic disc
C4	- 47µF, 16V electrolytic
C6	- 470µF, 25V electrolytic
Miscellaneous:	
S1-S2	- SPDT foot switch
S3	- Tactile switch
CON1	- 2-pin terminal connector
DIS1	- LTS543 7-segment display

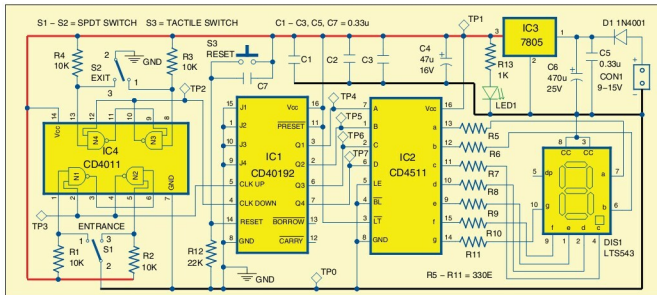


Fig. 1: Circuit diagram of the difference counter